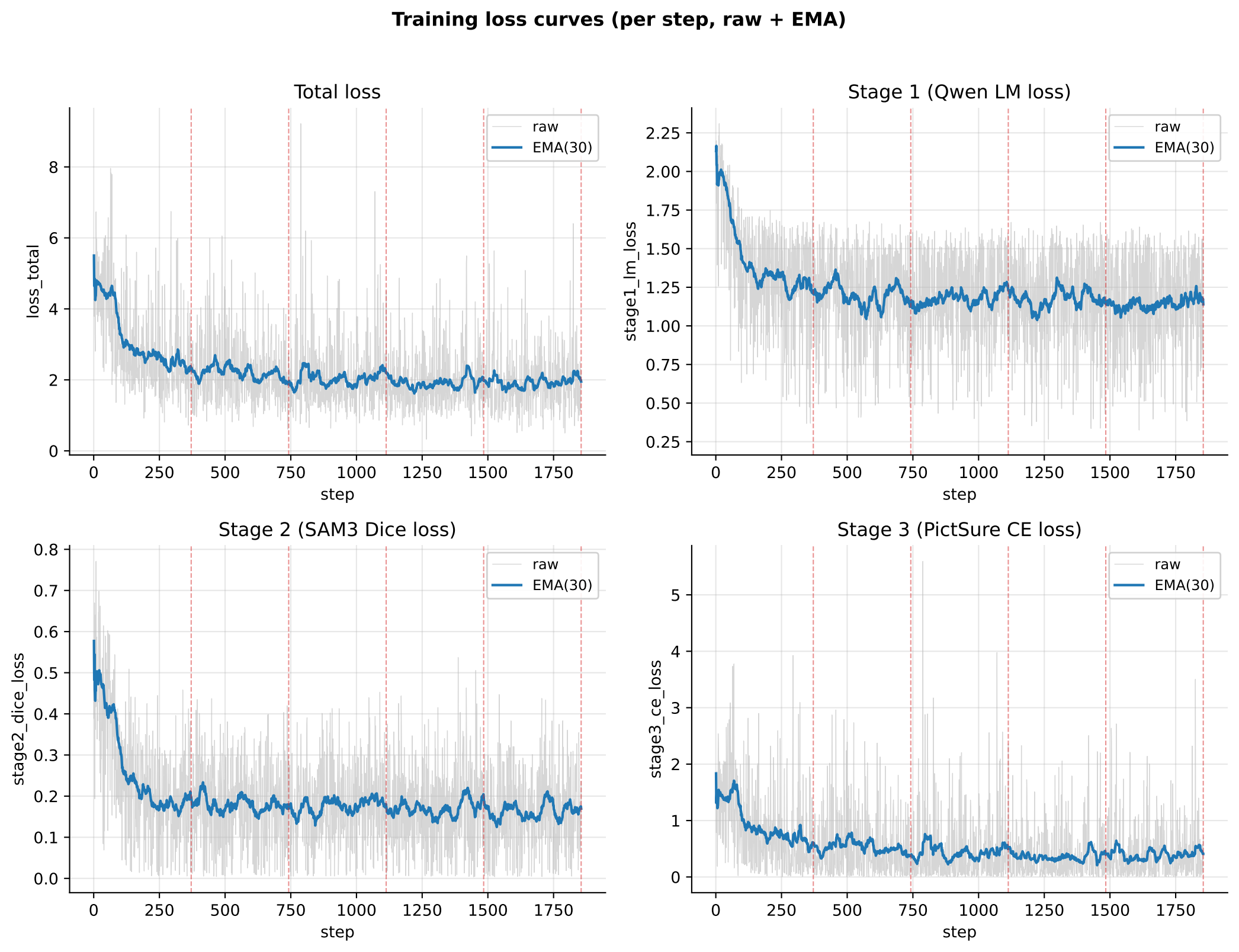
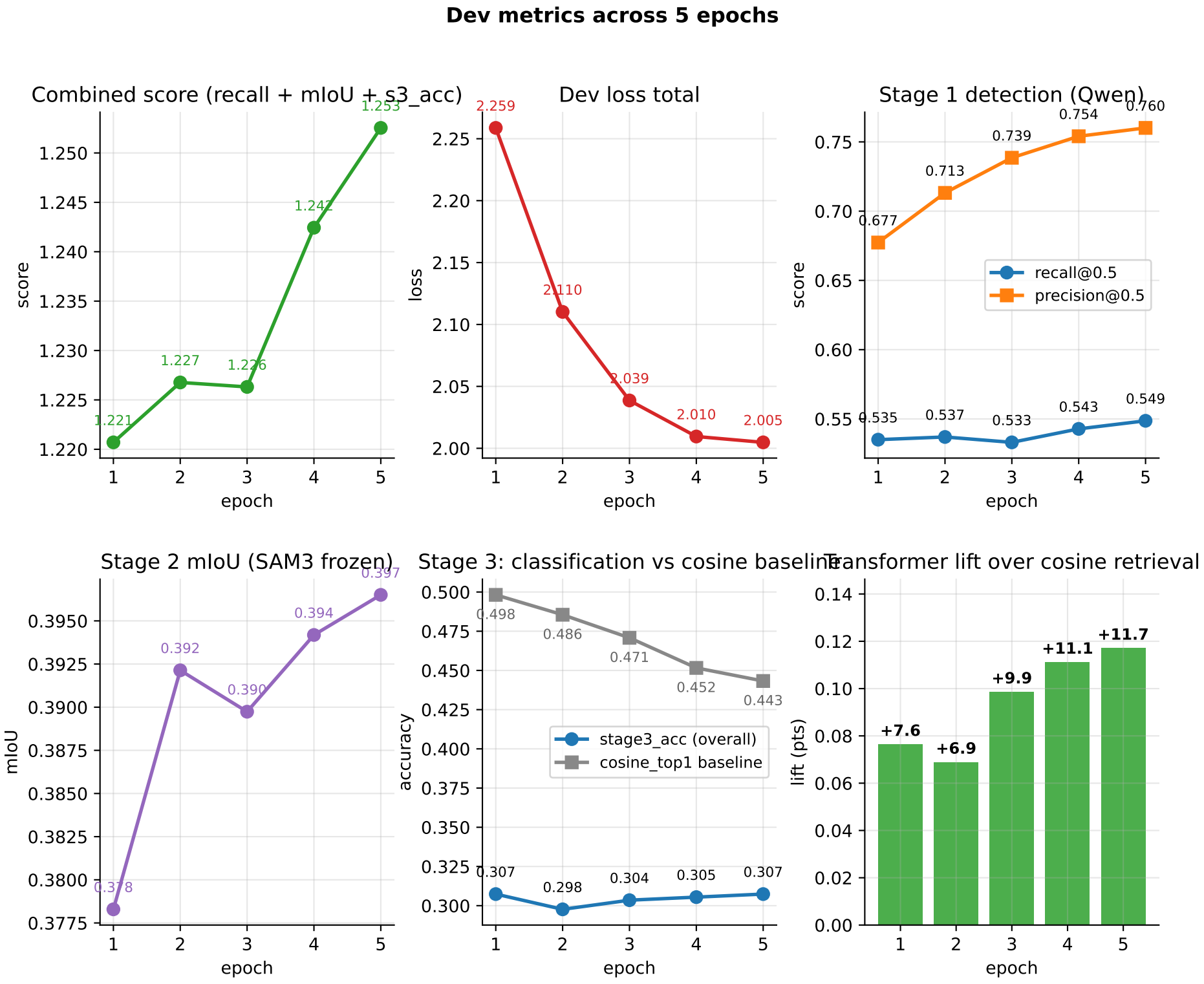
TriFoodNet Training Report

trial-20260425-2313-real-1hr

Notes	Best retained snapshot run packaged for resume/evaluation
Seed	1337
Determinism mode	deterministic
Run timestamp (UTC)	2026-04-25T23:19:39
Hardware	NVIDIA GeForce RTX 5090 (Blackwell sm_120)
Torch / CUDA	2.7.0+cu128 / cuda 12.8
transformers / peft	5.6.2 / 0.19.1
Image	pytorch/pytorch:2.7.0-cuda12.8-cudnn9-devel
Provider	vast.ai (Denmark, RTX 5090, $0.39/hr)
Dataset	v3 KFUPM food, hash 61ac038c
Splits	train 80% / dev 10% / test 10%
Dev split size	370 images / 514 items
Stage 1 (Qwen2.5-VL-3B)	LoRA r=16/α=32 on q/k/v/o (~7.4M trainable)
Stage 2 (SAM3)	FROZEN (image enc + prompt enc + mask dec)
Stage 3 (PictSure ViT)	ViT FROZEN, transformer LoRA r=16 (~524K trainable)
Total trainable / total params	7.90M / 4.72B (1 in 598)
Joint training epochs	5
Effective batch size	8 (bs=2 × grad_accum=4)
Per-stage LR	Qwen 2e-5 / SAM 5e-5 / PictSure 1e-4 (cosine schedule)
Teacher forcing	GT boxes throughout (use_gt_boxes=True)
Stage 1 chunk size (eval)	8 (batched Qwen.generate for ~5× faster dev eval)
Total wall time	253.8 min (4.23 hrs)
Dev evals completed	5 (one per epoch)
Best epoch (lowest dev_loss)	5 — dev/loss_total = 2.0048
Best combined score	1.2525


Training loss curves (per step, raw + EMA)
Total loss
loss_total
step
raw
EMA(30)
Stage 1 (Qwen LM loss)
stage1_lm_loss
step
raw
EMA(30)
Stage 2 (SAM3 Dice loss)
stage2_dice_loss
step
raw
EMA(30)
Stage 3 (PictSure CE loss)
stage3_ce_loss
step
raw
EMA(30)


Dev metrics across 5 epochs
Combined score (recall + mIoU + s3_acc)
score
epoch
1.221
1.227
1.226
1.242
1.253
Dev loss total
loss
epoch
2.259
2.110
2.039
2.010
2.005
Stage 1 detection (Qwen)
score
epoch
recall@0.5
precision@0.5
0.677
0.713
0.739
0.754
0.760
0.535
0.537
0.533
0.543
0.549
Stage 2 mIoU (SAM3 frozen)
mIoU
epoch
0.378
0.392
0.390
0.394
0.397
Stage 3: classification vs cosine baseline
accuracy
epoch
stage3_acc (overall)
cosine_top1 baseline
0.498
0.486
0.471
0.452
0.443
0.307
0.298
0.304
0.305
0.307
Transformer lift over cosine retrieval
lift (pts)
epoch
+7.6
+6.9
+9.9
+11.1
+11.7

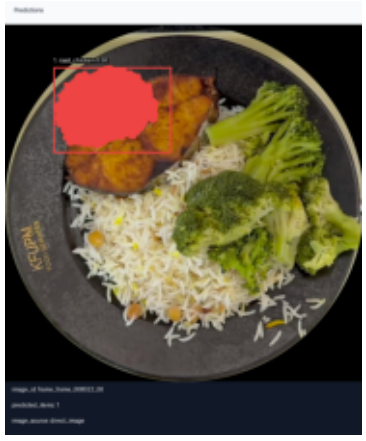
Full dev metrics — all 5 epochs

Metric	ep1	ep2	ep3	ep4	ep5	Δ
combined	1.2207	1.2268	1.2263	1.2424	1.2525	+0.0318
loss_total (dev)	2.2588	2.1102	2.0386	2.0095	2.0048	-0.2540
loss_stage1	1.2083	1.1729	1.1614	1.1580	1.1571	-0.0512
loss_stage2	0.4125	0.3939	0.3888	0.3881	0.3881	-0.0244
loss_stage3	0.5628	0.4936	0.4553	0.4383	0.4358	-0.1270
Stage 1: recall@0.5	0.5350	0.5370	0.5331	0.5428	0.5486	+0.0136
Stage 1: precision@0.5	0.6773	0.7132	0.7385	0.7541	0.7601	+0.0828
Stage 1: pred_items_per_image	1.0973	1.0459	1.0027	1.0000	1.0027	-0.0946
Stage 2: mIoU	0.3783	0.3921	0.3897	0.3942	0.3965	+0.0182
Stage 2: BCE loss	0.1130	0.1083	0.1082	0.1088	0.1087	-0.0043
Stage 2: Dice loss	0.1780	0.1699	0.1673	0.1668	0.1669	-0.0111
Stage 3: acc (overall)	0.3074	0.2977	0.3035	0.3054	0.3074	+0.0000
Stage 3: acc (of detected)	0.5745	0.5543	0.5693	0.5627	0.5603	-0.0143
Stage 3: episode acc (5w-5s)	0.8054	0.8268	0.8346	0.8405	0.8444	+0.0389
Stage 3: cosine top-1 acc	0.4982	0.4855	0.4708	0.4516	0.4433	-0.0549
Stage 3: acc when retrieval got GT	1	1	1	1	1	+0
Stage 3: retrieval recall@K	0.8473	0.8188	0.8248	0.8136	0.8191	-0.0281
Stage 3: TRANSFORMER LIFT	0.0764	0.0688	0.0985	0.1111	0.1170	+0.0407
Counts: n_images	370	370	370	370	370	+0
Counts: n_items (GT)	514	514	514	514	514	+0
Counts: n_pred_items	406	387	371	370	371	-35
Counts: n_matches (TP)	275	276	274	279	282	+7
Counts: n_with_candidates	275	276	274	279	282	+7
Counts: n_retrieval_hits	233	226	226	227	231	-2
Counts: n_cosine_top1_correct	137	134	129	126	125	-12
Latency: stage1 ms/img	1198.8	1128.0	1010.0	1022.9	1032.2	-166.6
Latency: stage2 ms/img	99.0	102.8	103.0	103.7	103.3	+4.3
Latency: stage3 ms/img	37.2	36.8	35.3	36.0	35.3	-1.9
Latency: TOTAL ms/img	1334.9	1267.6	1148.3	1162.6	1170.7	-164.2

Sample dev predictions (epoch 3) — 370 total saved

frame_frame_000049_00

frame_frame_008022_00



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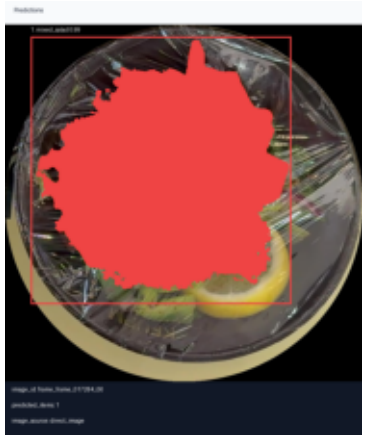
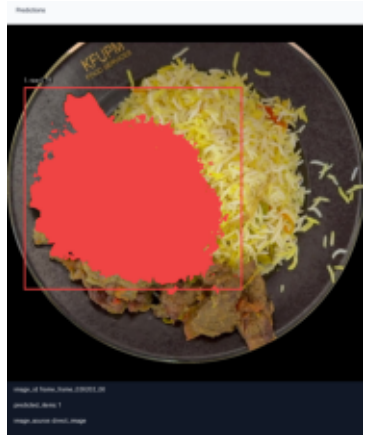
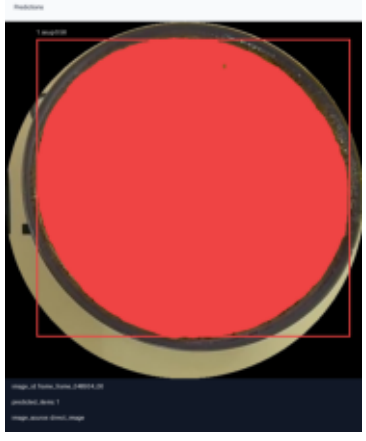


frame_frame_026202_00

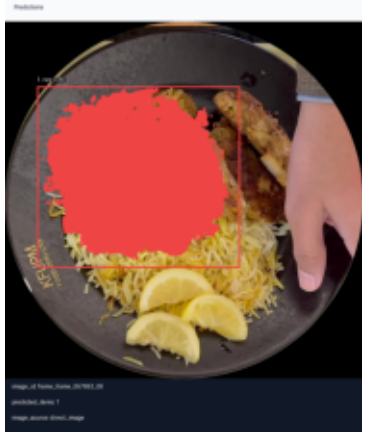


frame_frame_039273_00

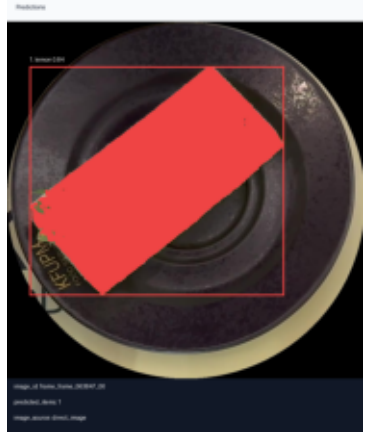
frame_frame_048934_00



frame_frame_057692_00

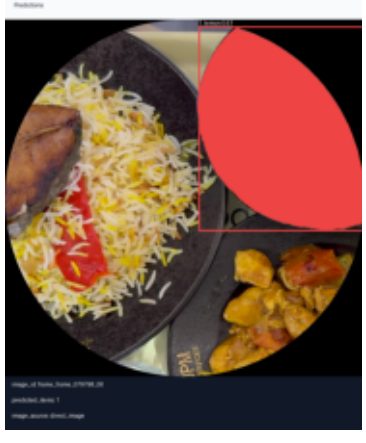


frame_frame_063847_00



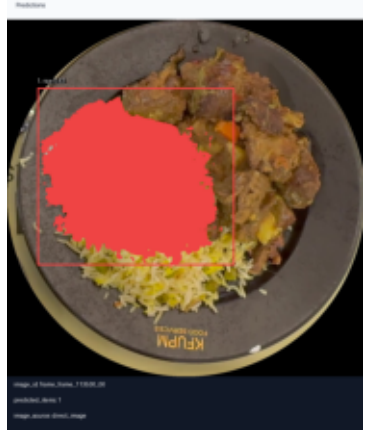
frame_frame_072290_00

frame_frame_079798_00



frame_frame_099361_00

frame_frame_113530_00



Key findings

Stage 1 (Qwen detection) — got smarter, not more

- Recall@0.5: 0.535 → 0.549 (Δ +0.014, essentially flat)
- Precision@0.5: 0.677 → 0.760 (Δ +0.083, +8.3 pt)
- Predictions made: 406 → 371 (Δ -35)
- Correct detections: 275 → 282 (Δ +7)
  ↳ Qwen learned to be conservative: fewer total predictions, higher hit rate.

Stage 2 (SAM3 mask) — free-rode on Stage 1

- mIoU: 0.378 → 0.397 (Δ +0.018)
  ↳ SAM3 is fully frozen. All gain is downstream: better Stage 1 boxes → cleaner SAM prompts → better masks.

Stage 3 (PictSure ICL) — the headline result

- Cosine top-1 baseline: 0.498 → 0.443 (Δ -0.055, -5.5 pt)
- Transformer lift: +7.6 pt → +11.7 pt (Δ +4.1 pt)
- Stage 3 acc (overall): 0.307 → 0.307 (Δ +0.000, FLAT)
- Episode acc (in-distribution 5w-5s): 0.805 → 0.844 (Δ +0.039)
  ↳ Joint training degrades the embedding's cosine geometry, but the transformer learns to compensate — gap widens monotonically.

Combined picture

- Combined score: 1.2207 → 1.2525 (Δ +0.0318)
- Dev loss total: 2.2588 → 2.0048 (Δ -0.2540)
- Latency: 1335 → 1171 ms / image
  ↳ Combined gains driven entirely by Stage 1 + Stage 2; Stage 3 acc is held flat by the embedding-degradation / transformer-compensation balance.

Bottleneck

- Stage 1 recall is the binding constraint: 45 % of GT items are never detected and therefore can't be classified. Halving stage3_acc gains is impossible without lifting recall.
- Path forward: train Stage 1 longer / with detection-specific loss / larger backbone, before further joint Stage 3 work.

Publishable claim

"On a 32-class restaurant food dataset, an ICL transformer adds +11.7 pt over pure cosine retrieval, and the gap widens monotonically with joint training (+7.6 pt at epoch 1 → +11.7 pt at epoch 5). This holds despite the cosine baseline degrading by -5.5 pt over the same training, indicating the transformer learns to extract more signal from the (jointly-tuned) embeddings."